Is Training in hot weather a more efficient way to train?

An analysis of Heat Acclimatization using the EndoMondo Fitness Dataset

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DATA301

1 Project Proposal

1.1 Summary

For my project I will be using the Endo Mondo fitness tracking dataset collected by Ni, Muhlstein and McAuley (2019). This dataset contains 253,020 workout logs from 1,104 different users. The dataset includes heart rate, speed, GPS, distance, sport type, gender and weather conditions recorded during each workout. My research question for the project is “How long does it take for regular hot weather training to produce measurable improvements in heart rate efficiency, and does this threshold differ across sport types such as running, cycling and hiking?” To answer this I will use Dask to process the data, group users by the weather conditions they typically train in, and then compare how their heart rate efficiency changes over time across those groups. I expect to find that users who consistently train in hot weather will show a gradual improvement in heart rate efficiency over time as their body gets used to the heat, while users who train in cooler conditions will show less improvement. This matters practically as it could give athletes and coaches data-driven evidence for whether heat training actually makes you fitter, and could also be used to improve personalised fitness recommendation apps.

1.2 Motivation

I have always had a love for sports and spend most of my free time training for triathlons. I recently spent that summer training in Australia which is a lot hotter climate than NZ. During this time I experienced firsthand how the weather can affect a workout. Training in the summer heat feels punishing to start with but I found that after a while my body adapted and it became more manageable. This made me curious to see if this observation is actually measurable in performance data and how long it takes to see these improvements. The findings could be useful for both coaches and athletes that are racing or training in hotter climates so that they can tailor their training respectively. The findings could also be used in fitness apps to provide more personalised workout recommendations depending on where the users workout history and the climate they live in.

1.3 Background

The Endo Mondo dataset contains 253,020 workouts from 1,104 different users, with every session recording sensor data including heart rate, speed, GPS coordinates, altitude, and distance, as well as different metadata including sport type, gender, and weather conditions. The weather field shows the environment during each workout and will help enable analysis of how the training environment relates to performance over time. One of the main metrics that I will be looking at to judge the increase in performance is the relationship between heart rate and the speed achieved during the exercise, this is called heart rate efficiency. A lower heart rate at the same speed shows an improved fitness and performance. Heat acclimatisation is a concept in sports science that talks about the adaptations that the body makes in response to training in heat. The algorithms that I will use in this project will be cosine similarity to cluster users by the climate they train in and time series analysis using Dask to track their heart rate efficiency changes across workouts and find where the improvement becomes measurable and significant. I will also separate the analysis for the different sports to see if the effects differ.

1.4 Research Question

The research question for my project is: “How long does it take for regular hot weather training to produce measurable improvements in heart rate efficiency, and does this threshold differ across sport types such as running, cycling and hiking?” This is a comparative, relationship based research question that looks into performance changes over time in response to the training climate, it then compares that change across different users and sport types. The dataset suits this question as it provides both personalised weather conditions and physiological metrics for the same user over a long period so we can track the performance over time not just have a small single snapshot. Cosine similarity will be used to find the clusters of users with similar training climates and the time series analysis will be used to track the heart rate efficiency across different hot weather workouts to find at what point we will be able to see the improvement. By repeating this analysis for running, cycling and tramping will test to see if the threshold differs across sports.

1.5 Design and Methods

My project will have several steps in its pipeline, all implemented in python using Dask to make the data processing scalable. First I will load the raw data into Dask and the workouts with missing data will be filtered out and heart rate efficiency will be added as a metric for every workout session. I will then separate the users by their training climate (hot and cool) and use cosine similarity to compare the users in hot climates to the users in cold climates. For the users in hot climates I will order the workouts in order of date recorded to calculate when the performance becomes visible. I will then redo this separately for every sport. Some of the difficulties that I may come across is that some users may have less workouts recorded than others so I will make a minimum number of workouts required and exclude any users with less than that number. I will also have to consider different variables such as gender, and workout duration to make sure these don’t affect my results.

1.6 References

Ni, J., Muhlstein, L., & McAuley, J. (2019). Modeling heart rate and activity data for personalized fitness recommendation. WWW.

Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). Mining of Massive Datasets (3rd ed.). Cambridge University Press.

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Sample Project Notebook. (2024). DATA301 sample project code. Provided by course instructor.

2 Progress Report

2.1 What have you done so far on your project?

So far in my project I have loaded in the EndoMondo dataset into python using pandas, gzip and ast. I loaded in the metadata file and removed any rows/entries of data that didn't contain any weather data. I then created a data frame for me to work with containing variables such as user ID, sport type, weather, duration, distance, calories, and workout date.

I then explored my data by looking into how many total workouts and unique users I have. I then looked into the most common weather types that people recorded and the most common sports recorded. I also checked to see if there was any missing values and what the date range was between the oldest and most recent data.

I then took to cleaning up the data by converting the time date format to a correct more usable format. I then merged bike and bike transport into one sport as they are both similar enough that I will be able to compare all of their HR performance data as if it was all recorded as the same sport. The dataset contains a lot more sports than I want to look at so after merging bike and bike transport I removed all the sports apart from bike, run and walk as these are the sports that I want to study. They are also the sports with the most amount of recorded activities attached to them. I then removed all users with less than 20 workouts in any of these sports as in the study I am looking at performance over time so no use at looking at someone who has just worked out a couple of times.

2.2 What are you doing next?

Next I plan on calculating a heart rate efficiency metric that I can apply to each workout. I will then group users based on the climate that they usually train in. After this, I will use time series analysis to track how heart rate efficiency changes over time for users who usually train in hot weather vs users who usually train in cooler weather. I also plan on repeating the analysis for the different sports to see if heat acclimatisation applies to one sport more than the others. After this I will make some visualisations and report on my findings.

2.3 Do you have any roadblocks?

Some roadblocks that I have encountered is that the weather data is categorical rather than the expected numerical, this means I will have to be careful on how I define what "hot" weather means and will probably require some further research. Another roadblock I will have to work through is being able to consistently measure heart rate efficiency across sports as each sport has different intensities.

Final Report

Abstract

For this project I used the EndoMondo fitness tracking dataset collected by Ni, Muhlstein and McAuley (2019). The dataset contains 253,020 workout logs from 1,104 users, it includes data such as heart rate, speed, distance, sport type, gender and weather conditions. My research question was: “Do users who regularly train in hot weather conditions show improved heart rate efficiency over time compared to users who train in cooler conditions?” To answer this I used Python and Dask to process the data and several other techniques such as user clustering, rolling average analysis and set operations to compare the performance of people training in hot weather vs cool weather. I found that 95.8% of hot weather trainers showed improvement in heart rate efficiency (heart rate vs speed) compared to 77.8% of cool weather trainers. I also found that 94 users improved specifically in hot weather but not cool weather. These results directly support the idea that heat training produces better improvement. This is relevant for athletes and coaches who are preparing for competition in hot climates and want true evidence for whether heat training actually works.

3 Introduction

3.1 Background

The EndoMondo dataset contains workout logs recorded from July 2012 up until April 2019. The dataset comes in two files. The first one is called endomondoHR.json.gz, and it contains the heart rate and speed data recorded during each workout. The other file is called endomondoMeta.json.gz and contains summary information such as weather, sport type and duration. Weather conditions are stored as numerical codes from AccuWeather ranging from clear sunny conditions through to rain, snow and fog. In this project to see the gain in fitness I used a metric called heart rate efficiency. Heart rate efficiency is calculated by dividing average speed by average heart rate. A higher score means the athlete is moving faster for less cardiovascular effort, indicating better fitness.

In my project I was looking to see if the well known concept of heat acclimatisation has positive effects on heart rate efficiency. Heat acclimatisation is the process that the human body goes through when repeatedly exposed to heat during exercise. The body learns it needs to adapt to this heat to keep efficient. It can lead to a better sweating responses, better lung performance and lower heart rate (Périard, Racinais and Sawka, 2015). Research and recent studies suggest that these adaptations start to appear after around seven to fourteen days of consistent heat exposure while training.

3.2 Motivation

I grew up doing triathlon and ended up competing at a high level during high school and university, leading to racing in several different climates across the world. Due to these races and living and training in Australia last summer I have experienced firsthand how much the weather can affect a workout. Training in the summer heat feels really hard at first, but after a while the body adapts and it becomes more manageable. When I came back from living in the hot Australia climate to the cool

New Zealand weather I noticed that I felt a lot fitter than when I left. This made me curious whether the adaptation I noticed in my own training is actually measurable in real data. As said previously, the findings could be useful for athletes and coaches planning training blocks before hot weather races, and for fitness apps that want to give more personalised recommendations based on where and when users train.

3.3 Research Question

My research question is: "Do users who regularly train in hot weather conditions show improved heart rate efficiency over time compared to users who train in cooler conditions?" This is a comparative research question that looks at how performance changes over time depending on training environment/weather conditions. The EndoMondo dataset suits this question well because it has both weather data and physiological performance data for the same users over several years. This means that I can track peoples fitness over time and see whether the weather they train in plays a role. The algorithms I used, rolling averages, clustering and set operations directly help answer this by tracking performance trends and identifying which users improved specifically in hot versus cool conditions.

4 Experimental Design and Methods

During the project I ran into a few challenges. The main challenge was how to efficiently load in the dataset. Originally I was loading the data directly from the compressed files using "gzip" and "ast.literal_eval". Both files are compressed and stored in a format that requires processing them line by line. Sometimes this could take up to 55 minutes per file on my laptop. This made it really hard to test and develop the code because every small change meant waiting ages to reload the data.

After a recommendation from a mate I chose to pre-process both files into clean CSV files using a separate python script. Then I uploaded those CSVs to Google Drive. From there I could download them in the notebook using the "gdown" library. This made the whole development process way faster. In the preprocess.py script, I loaded in the meta data first making sure to filter out any workouts that had missing weather data as workouts without it were useless to my project. I then saved the result as a cleaned CSV. Before loading in the heart rate file I used the metadata workout IDs to pre filter out any workouts that weren't in the metadata file for example data with no weather conditions attached. The script then cleans the heart rate data by removing any null values or unrealistic outliers that were most likely due to inaccurate equipment.

The main script starts by downloading and merging the two cleaned data files on workout ID. I then cleaned the data by converting dates to the right format, merging bike and bike transport into one category, filtering to only keep running, cycling and walking workouts, and removing any users with fewer than 20 workouts per sport. Each workout was then labelled as either hot or cool based on the AccuWeather weather codes. From there I ran four main analyses. First I found the most frequent item sets using a Dask Bag to find which sport and weather combinations were most common amongst users. I then made a rolling average of heart rate efficiency. This was calculated using Dask's parallel processing to compare how efficiency changed over time for hot weather workouts vs cool weather workouts. Third was clustering users into three groups, people who mainly trained in hot weather, people who mainly trained in a mix of both hot and cool weather and also people who did most of their training in cool weather. My fourth analysis was union, intersection and difference set operations to find which users improved specifically in hot weather, specifically in cool weather, or in both.

I also attempted to use cosine similarity to compare users based on what climate they train in. This was done with the goal of finding users who had the most similar training habits and trying to compare whether they showed similar performance improvements over time. However this approach did not produce results I was after, the user vectors only had three possible dimensions, hot running proportion, hot cycling proportion and hot walking proportion. This meant most users ended up with very similar vectors and similarity scores clustered around 1.000 regardless of their training patterns. The dataset simply did not have enough variation to make cosine similarity a useful way to differentiate the users, so clustering based on overall hot weather proportion was used instead as a better way to group users.

4.1 Libraries and Functions Used

- *gdown* — downloading the pre-processed CSV files from Google Drive
- *pandas* — data loading, cleaning and merging
- *dask.dataframe* — parallel processing using *map_partitions*
- *dask.bag* — frequent itemsets using *get_pairs*, *.map()*, *.flatten()*, *.frequencies()* and *foldby*
- *matplotlib* — plotting graphs

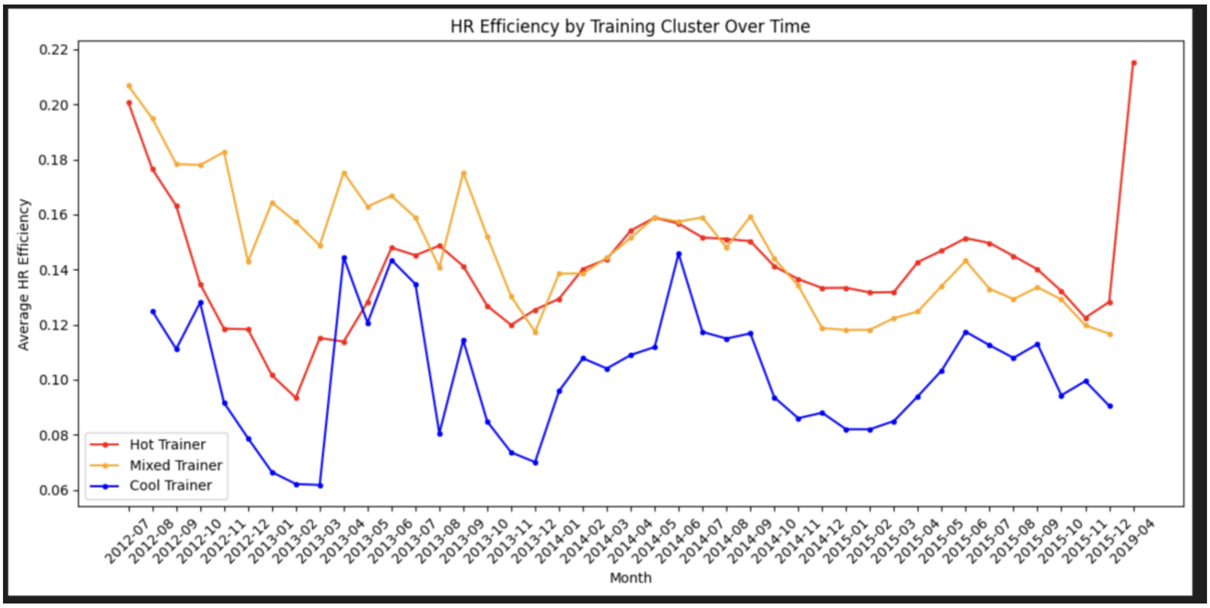
- *get_pairs(basket)* — finds all sport and weather combinations pairs per user
- *compute_rolling_efficiency(df)* — calculates a 10 workout rolling average of heart rate efficiency per user
- *cluster_user(proportion)* — assigns each user to a hot, mixed or cool trainer cluster
- *did_user_improve(workouts)* — compares the first five and last five workouts to check if a user improved

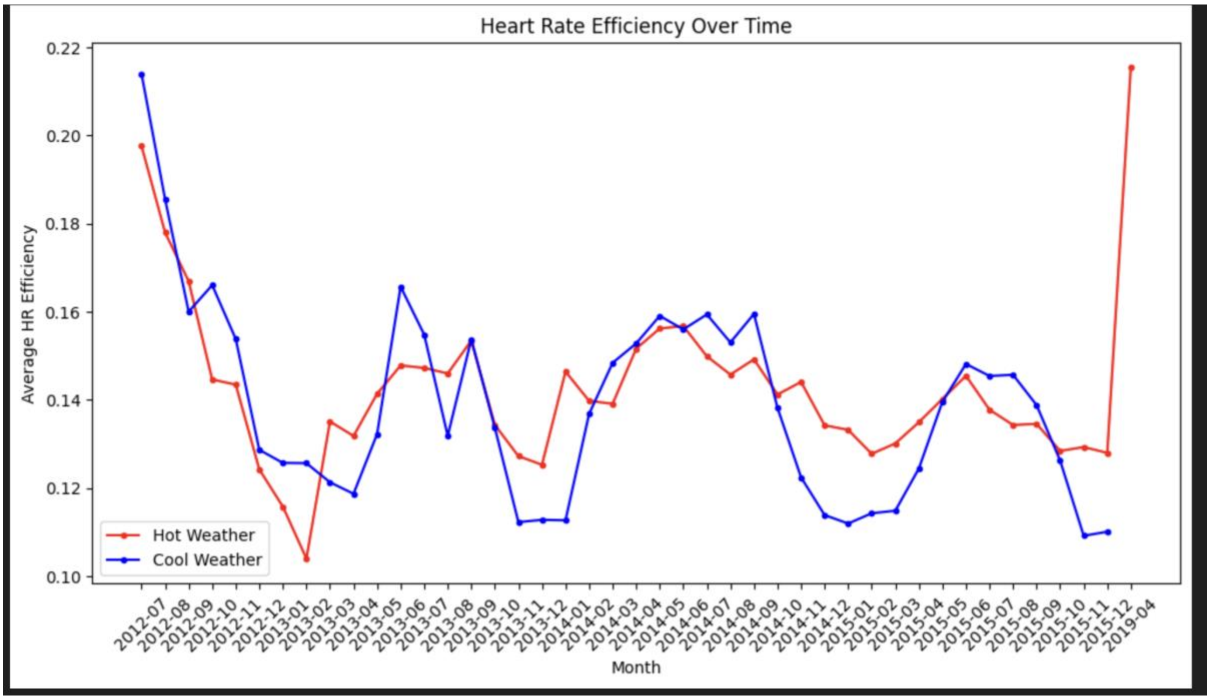
5 Results

After cleaning the data I had 38,988 workouts across 855 unique users with running being the most popular sport in the dataset with 18,064 workouts recorded spanning from July 2012 to April 2019. After filtering out users with less than 20 workouts per sport I had 35,111 workouts left.

The frequent itemset analysis showed that the most common combination in the dataset was cool running paired with hot running with 249 users appearing in this category. The next most common was cool cycling paired with hot cycling with 222 users. This tells us that most users train in a combination of both hot and cold weather rather than training just in one. This is important as it means we are mainly comparing the same people doing the same sport but in different weather conditions, not just 2 completely separate groups of people who only train in one environment. This makes the comparison and our findings more accurate and meaningful because it means that any difference in performance we see will more likely be due to weather conditions actually having an effect rather than one group just containing “fitter” people in general.

The rolling average graph showed that hot weather workouts consistently produced higher heart rate efficiency than cool weather workouts across most of the dataset. The clustering graph backed this up as the hot trainer cluster sat noticeably above the cool trainer cluster throughout the entire observation period, with cool trainers showing particularly low efficiency from 2014 onwards.





The union, intersection and difference analysis gave the strongest results. 95.8% of hot weather trainers showed improvement in heart rate efficiency over their workout history, compared to 77.8% of cool weather trainers. Also 94 users improved specifically in hot weather but not in cool weather, compared to only 15 users who improved specifically in cool weather but not hot. That is more than a six to one ratio in favour of hot weather improvement. This gives overall strong evidence that hot weather training produces adaptations that cool weather training does not.

6 Conclusion

The results clearly support the hypothesis that hot weather training leads to better improvements in heart rate efficiency than cool weather training. Nearly all users who train in hot weather showed improvement over time. A significant group of 94 users improved specifically in hot conditions but not cool ones. This suggests the heat itself is driving the increase in performance rather than just general fitness gains.

For athletes and coaches these findings are highly useful. It shows that a hot weather training block can produce real improvements in performance. For athletes that are preparing for races or events in hot climates it shows clear data driven support for including heat training in their preparation. Fitness apps could also use this kind of analysis to give more personalised recommendations, users training for hot weather events could be guided toward heat training blocks based on their workout history.

If I did future research, it would look at controlling other variables like workout intensity and duration as these also have an effect on performance but were not accounted for in this project. It would also be interesting to investigate exactly how many hot weather sessions are needed before improvement becomes measurable, which was part of the original research question but proved too complex to answer with this dataset. Using a more recent dataset would also be valuable as training habits and wearable technology have changed a lot since 2019 and would be likely to give us more accurate results and data.

7 Critique of Design and Project

The biggest limitation of my approach was classifying all weather conditions into just two categories of hot or cool. This loses a lot of detail because a mild sunny day and an extremely hot day both get labelled as hot, even though they would produce very different physiological stresses. A better approach would have been to use actual temperature values. This would have allowed for a better analysis of how performance changes across a range of temperatures rather than just two groups. Another issue was that the rolling average was calculated across all workouts regardless of weather, leading to a user's efficiency score at any point reflecting their overall fitness rather than their fitness specifically in hot or cool conditions. Calculating separate rolling averages for hot and cool workouts per user would have been more accurate but also more complex to implement.

8 Reflection

The most useful things from the course for this project were Dask's parallel processing, Dask Bag's map and foldby operations, and frequent itemset mining. The lab exercises were especially useful for the get_pairs, .flatten() and .frequencies() functions used. The biggest practical lesson I learned was how much data preprocessing matters. The 55 minute load time for the raw files was a massive obstacle. Solving it with pre-processed CSVs and Google Drive was the most impactful change I made to the whole project. I also learned that choosing the right metric is a really important step. Heart rate efficiency turned out to be a simple but effective measure that captured real performance differences.

References

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Jasper Sharp gave advice on how to make the data loading processes quicker.

AI, specifically Claude, was used to help format references and spell/grammar check the written work. When developing the code, Claude was used to help find and teach relevant Dask and pandas documentation/functions. It was also used for debugging in helping interpret error messages.

Project Code

preprocess.py

```
import gzip
import ast
import pandas as pd



# Processing meta data file

meta_rows = []
# open the compressed file
with gzip.open('endomondoMeta.json.gz') as f:
    for i, line in enumerate(f):
        # convert lines into dictionaries
        row = ast.literal_eval(line.decode('utf-8'))
        # skip row if weather data is missing
        if row['weather'] is None:
            continue
        # put data fields into a clean format
        meta_rows.append({
            'user_id':   row['userId'],
            'workout_id': row['id'],
            'sport':     row['sport'],
            'gender':    row['gender'],
            'weather':   row['weather']['type'],
            'date':      row['timestamp'][0],
            'duration':  row['duration'],
            'distance':  row['distance'],
            'calories':  row['calories']
        })

# convert dictionaries into a dataframe
meta_df = pd.DataFrame(meta_rows)
# save file
meta_df.to_csv('meta_cleaned.csv', index=False)

# take workout IDs to filter HR dataset
needed_ids = set(meta_df['workout_id'].values)



# Processing hr file

hr_rows = []
with gzip.open('endomondoHR.json.gz') as f:
    for i, line in enumerate(f):
        row = ast.literal_eval(line.decode('utf-8'))
        # filter out rows that are not in the new cleaned meta data file
        if row['id'] not in needed_ids:
            continue
        heart_rate = row.get('heart_rate')
        speed      = row.get('speed')
        # exclude if missing hr or speed data
        if not heart_rate or not speed:
            continue
        hr_rows.append({
            'workout_id':     row['id'],
            'avg_heart_rate': sum(heart_rate) / len(heart_rate),
            'avg_speed':      sum(speed) / len(speed),
        })

hr_df = pd.DataFrame(hr_rows)
```

```
# calculate hr efficiency metric
hr_df['hr_efficiency'] = hr_df['avg_speed'] / hr_df['avg_heart_rate']
hr_df = hr_df[hr_df['avg_heart_rate'] > 0]
hr_df = hr_df[hr_df['avg_speed'] > 0]
hr_df = hr_df[hr_df['hr_efficiency'] != float('inf')]
hr_df = hr_df[hr_df['hr_efficiency'] < 10]

# save cleaned dataset
hr_df.to_csv('hr_cleaned.csv', index=False)

print("\nDone!")
```

project.ipynb

```
import gdown
import pandas as pd

# download pre-processed files from google drive
gdown.download("https://drive.google.com/uc?id=1XeC5mfGaIheTIEKwhmLRR--7QDVWONwk",
               "meta_cleaned.csv", quiet=False)
gdown.download("https://drive.google.com/uc?id=1if4i8GUvCJGObl7fi0plA9fjTznkp2Vl",
               "hr_cleaned.csv", quiet=False)

# load both csvs
df    = pd.read_csv("meta_cleaned.csv")
hr_df = pd.read_csv("hr_cleaned.csv")

# merge on workout id
df = df.merge(hr_df, on="workout_id", how="inner")

print(f"Loaded {len(df)} workouts")
print(f"Unique users: {df['user_id'].nunique()}")
print(df.head())
```

```
Downloading...
From: https://drive.google.com/uc?id=1XeC5mfGaIheTIEKwhmLRR--7QDVWONwk
To: /Users/maxblockley/Library/CloudStorage/OneDrive-UniversityofCanterbury/uni26/data/project(1)/max_b_heat_acclimisation_project/meta_cleaned.csv
100%|██████████| 57.6M/57.6M [00:02<00:00, 21.1MB/s]
Downloading...
From: https://drive.google.com/uc?id=1if4i8GUvCJGObl7fi0plA9fjTznkp2Vl
To: /Users/maxblockley/Library/CloudStorage/OneDrive-UniversityofCanterbury/uni26/data/project(1)/max_b_heat_acclimisation_project/hr_cleaned.csv
100%|██████████| 2.28M/2.28M [00:00<00:00, 4.38MB/s]
Loaded 38988 workouts
Unique users: 855
    user_id  workout_id          sport gender  weather  \
0  10014612   322483848  mountain bike   male        6
1  10014612   322484176  mountain bike   male        7
2  10014612   322484222  mountain bike   male        7
3  10027353   614308474           bike   male        3
```

```
4  10027353   444987383         bike   male       7

                    date   duration    distance  calories  avg_heart_rate  \
0  2014-04-07T17:18:13.000Z   1904.690   10.763090     424.0      144.698000
1  2013-12-22T11:36:29.000Z   1694.520    7.087470     408.0      156.642000
2  2013-11-21T20:00:04.000Z   5157.140   29.636370     833.0      137.660000
3  2015-10-06T17:56:15.000Z  12978.237   96.957778    1489.0      115.432866
4  2014-12-05T18:18:35.000Z  12386.545  102.361920    1652.0      126.781563

   avg_speed  hr_efficiency
0  21.133598       0.146053
1  15.946315       0.101801
2  24.405955       0.177292
3  25.165003       0.218006
4  29.800552       0.235054
```

```
# Explore data

print(f"Total Workouts: {len(df)}")
print(f"Unique Users: {df['user_id'].nunique()}")
print(f"Weather type counts: {df['weather'].value_counts().head(10)}")
print(f"Sport Types: {df['sport'].value_counts().head(5)}")
print(f"Missing values:")
print(df.isnull().sum().head(5))
print(f"\nDate range--")
print(f"Earliest: {df['date'].min()}")
print(f"Latest: {df['date'].max()}")
```

```
Total Workouts: 38988
Unique Users: 855
Weather type counts: weather
1     7849
3     6922
6     5065
7     4212
4     2776
33    2223
2     1780
11    1473
35    1399
38    1319
Name: count, dtype: int64
Sport Types: sport
run              18064
```

```
bike              17248
bike (transport)    1185
mountain bike        986
indoor cycling       725
Name: count, dtype: int64
Missing values:
user_id       0
workout_id    0
sport         0
...

Date range--
Earliest: 2012-07-17T16:22:48.000Z
Latest: 2019-04-07T00:46:30.000Z
```

```
# Clean up the data
df['date'] = pd.to_datetime(df['date'])

# merge bike and bike transport into one category
df['sport'] = df['sport'].replace('bike (transport)', 'bike')

# filter to only keep sports we want
sports_to_keep = ['run', 'bike', 'walk']
df = df[df['sport'].isin(sports_to_keep)]

# remove users with less than 20 workouts per sport
workouts_per_user_sport = df.groupby(['user_id', 'sport']).size()
combos_to_keep = workouts_per_user_sport[workouts_per_user_sport >= 20].index
df = df[df.set_index(['user_id', 'sport']).index.isin(combos_to_keep)]
```

```
Sample date: 2014-04-07 17:18:13+00:00

Cleaned dataframe sample:
    user_id  workout_id sport gender  weather                      date  \
10  10028940   557373470   run   male        7 2015-07-08 16:22:32+00:00
11  10028940   551135560   run   male        4 2015-06-28 11:45:51+00:00
12  10028940   501753192   run   male        6 2015-04-12 11:38:51+00:00
13  10028940   501753188   run   male       13 2015-04-12 10:19:01+00:00
14  10028940   500674621   run   male        1 2015-04-10 16:43:34+00:00

    duration  distance  calories  avg_heart_rate  avg_speed  hr_efficiency
10  3796.387  10.49797     668.0      133.510000  10.054001       0.075305
11  3869.402   8.06509     402.0      110.316000   7.612250       0.069004
12   504.876   1.22430     105.0      132.174242   8.538873       0.064603
```

13	4773.800	13.53379	769.0	145.440000	10.782886	0.074140
14	3749.502	11.00786	744.0	147.654000	10.700561	0.072471

```
# Label weather as hot or cool
hot_weather  = [1, 2, 3, 4, 33, 34, 35, 36]
cool_weather = [6, 7, 11, 12, 13, 14, 15, 16, 17, 18, 19, 22, 29, 38]

def label_weather(weather_type):
    if weather_type in hot_weather:
        return 'hot'
    elif weather_type in cool_weather:
        return 'cool'
    else:
        return 'unknown'

df['weather_label'] = df['weather'].apply(label_weather)
df = df[df['weather_label'] != 'unknown']
```

```
import dask.bag as db
import dask.dataframe as dd

# Frequent Itemsets
def get_pairs(basket):
    pairs = []
    for i in range(len(basket)):
        for j in range(i+1, len(basket)):
            pair = tuple(sorted((basket[i], basket[j])))
            pairs.append(pair)
    return pairs

baskets = df.groupby('user_id').apply(
    lambda x: list(set(x['weather_label'] + '_' + x['sport']))
).tolist()

baskets_bag = db.from_sequence(baskets)
all_pairs   = baskets_bag.map(get_pairs).flatten()
pair_counts = all_pairs.frequencies()
top_pairs   = pair_counts.topk(10, key=1).compute()

print("Top 10 most frequent sport/weather combinations:")
for pair, count in top_pairs:
    print(f"  {pair[0]} + {pair[1]}: {count} users")
```

baskets = df.groupby('user_id').apply(

Top 10 most frequent sport/weather combinations that appear together:

cool_run + hot_run: 249 users

cool_bike + hot_bike: 222 users

cool_bike + hot_run: 49 users

hot_bike + hot_run: 49 users

cool_bike + cool_run: 49 users

cool_run + hot_bike: 49 users

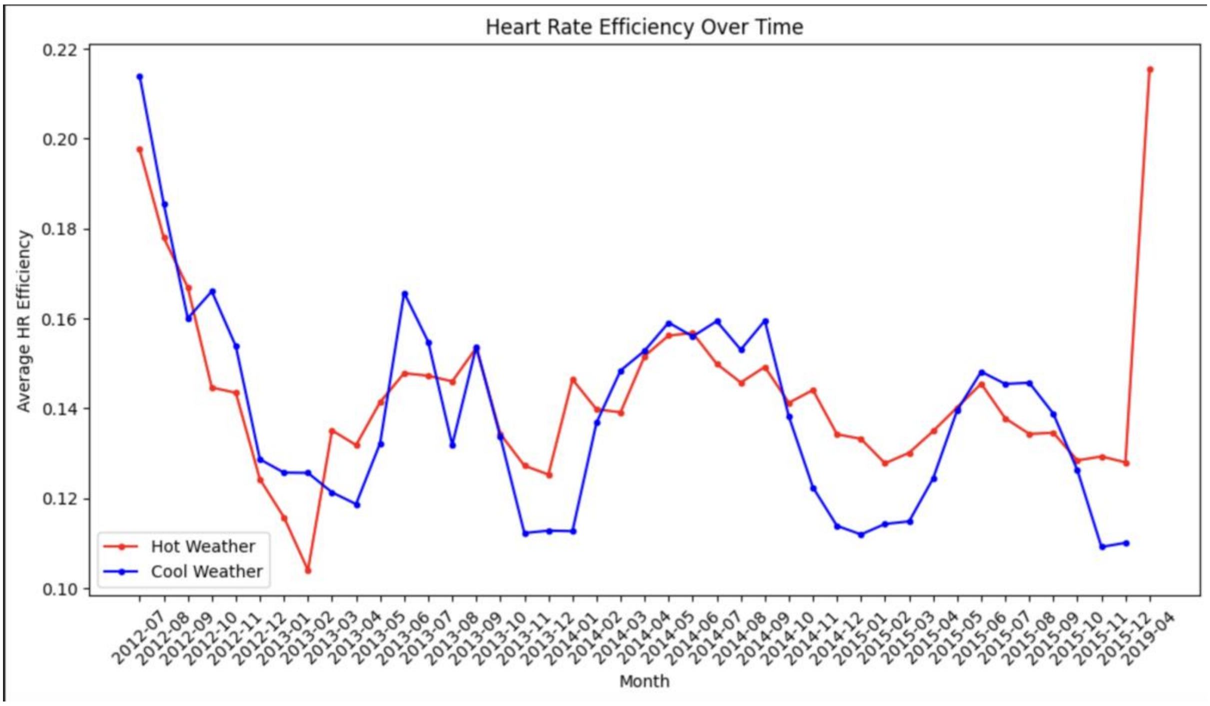
cool_walk + hot_walk: 6 users

hot_run + hot_walk: 5 users

cool_walk + hot_run: 5 users

cool_run + hot_walk: 5 users

```
# Rolling average of heart rate efficiency
ddf = dd.from_pandas(df, npartitions=4)

def compute_rolling_efficiency(df):
    df = df.sort_values('date')
    df['rolling_efficiency'] = df.groupby('user_id')['hr_efficiency'].transform(
        lambda x: x.rolling(window=10, min_periods=3).mean()
    )
    return df

df = ddf.map_partitions(compute_rolling_efficiency).compute()
df = df.dropna(subset=['rolling_efficiency'])

df['month'] = df['date'].dt.to_period('M').astype(str)
monthly = df.groupby(['weather_label',
'month'])['rolling_efficiency'].mean().reset_index()
monthly = monthly.sort_values('month')

# Plot HR efficiency: hot vs cool
import matplotlib.pyplot as plt

hot  = monthly[monthly['weather_label'] == 'hot']
cool = monthly[monthly['weather_label'] == 'cool']

plt.figure(figsize=(12, 6))
plt.plot(hot['month'],  hot['rolling_efficiency'],  label='Hot Weather',
color='red',  marker='o', markersize=3)
plt.plot(cool['month'], cool['rolling_efficiency'], label='Cool Weather',
color='blue', marker='o', markersize=3)
plt.xlabel('Month')
plt.ylabel('Average HR Efficiency')
plt.title('Heart Rate Efficiency Over Time')
plt.legend()
plt.xticks(rotation=45)
plt.show()
```




```
# Clustering users by hot weather training proportion
if 'cluster' in df.columns:
    df = df.drop(columns=['cluster'])

user_hot_proportion = df.groupby('user_id').apply(
    lambda x: (x['weather_label'] == 'hot').sum() / len(x),
    include_groups=False
).reset_index()
user_hot_proportion.columns = ['user_id', 'hot_proportion']

def cluster_user(proportion):
    if proportion >= 0.6:
        return 'hot trainer'
    elif proportion >= 0.4:
        return 'mixed trainer'
    else:
        return 'cool trainer'

user_hot_proportion['cluster'] =
user_hot_proportion['hot_proportion'].apply(cluster_user)
print("User clusters:")
print(user_hot_proportion['cluster'].value_counts())

df = df.merge(user_hot_proportion[['user_id', 'cluster']], on='user_id',
how='left')

print("\nAverage HR efficiency by cluster:")
print(df.groupby('cluster')['hr_efficiency'].mean())

cluster_monthly = df.groupby(['cluster',
'month'])['rolling_efficiency'].mean().reset_index()
cluster_monthly = cluster_monthly.sort_values('month')
```

```
hot_cluster   = cluster_monthly[cluster_monthly['cluster'] == 'hot trainer']
mixed_cluster = cluster_monthly[cluster_monthly['cluster'] == 'mixed trainer']
cool_cluster  = cluster_monthly[cluster_monthly['cluster'] == 'cool trainer']

plt.figure(figsize=(12, 6))
plt.plot(hot_cluster['month'],   hot_cluster['rolling_efficiency'],   label='Hot
Trainer',   color='red',    marker='o', markersize=3)
plt.plot(mixed_cluster['month'], mixed_cluster['rolling_efficiency'], label='Mixed
Trainer', color='orange', marker='o', markersize=3)
plt.plot(cool_cluster['month'],  cool_cluster['rolling_efficiency'],  label='Cool
Trainer',  color='blue',   marker='o', markersize=3)
plt.xlabel('Month')
plt.ylabel('Average HR Efficiency')
plt.title('HR Efficiency by Training Cluster Over Time')
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

User clusters:

cluster

hot trainer 249

mixed trainer 150

cool trainer 27

Name: count, dtype: int64

Average HR efficiency by cluster:

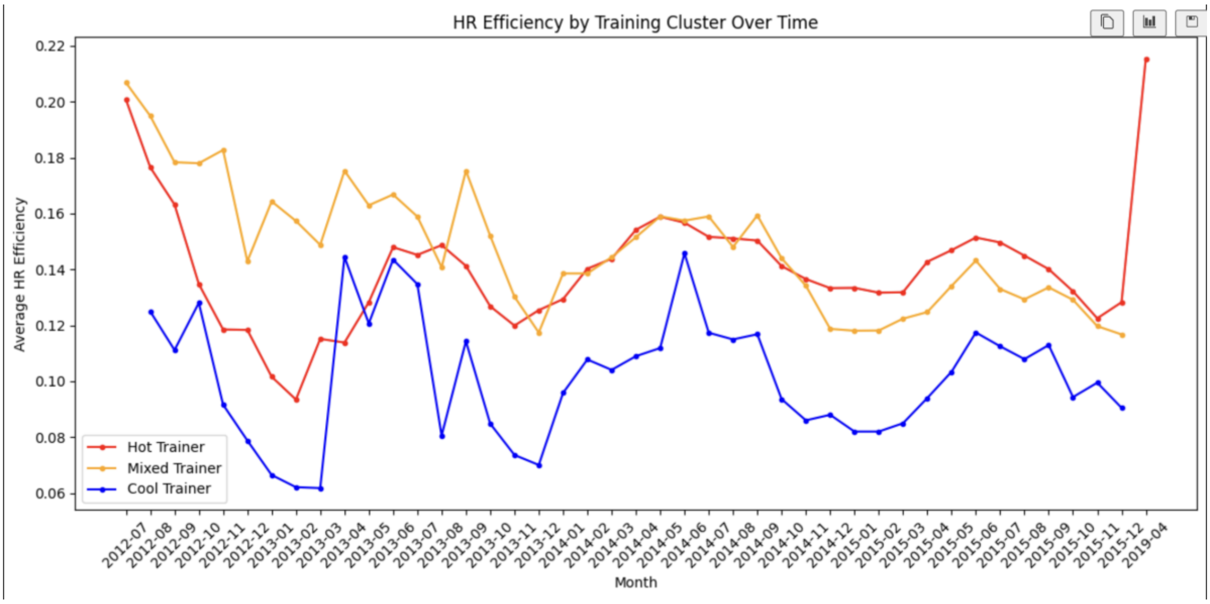
cluster

cool trainer 0.101472

hot trainer 0.142173

mixed trainer 0.139632

Name: hr_efficiency, dtype: float64



```
import dask.bag as db

# Union, Intersection, Difference — which users improved in which weather

def did_user_improve(workouts):
    workouts = sorted(workouts, key=lambda x: x['date'])
    if len(workouts) < 10:
        return None
    start = sum(w['rolling_efficiency'] for w in workouts[:5]) / 5
    end   = sum(w['rolling_efficiency'] for w in workouts[-5:]) / 5
    return (workouts[0]['user_id'], end > start)

hot_bag  = db.from_sequence(df[df['weather_label'] == 'hot'].to_dict('records'))
cool_bag = db.from_sequence(df[df['weather_label'] == 'cool'].to_dict('records'))

hot_grouped = hot_bag.foldby(
    key=lambda x: x['user_id'],
    binop=lambda acc, v: acc + [v],
    initial=[],
    combine=lambda a, b: a + b
)
cool_grouped = cool_bag.foldby(
    key=lambda x: x['user_id'],
    binop=lambda acc, v: acc + [v],
    initial=[],
    combine=lambda a, b: a + b
)

hot_results  = dict((uid, did_user_improve(workouts)) for uid, workouts in
hot_grouped.compute())
cool_results = dict((uid, did_user_improve(workouts)) for uid, workouts in
cool_grouped.compute())

hot_improvers  = set(uid for uid, improved in hot_results.items()  if improved)
cool_improvers = set(uid for uid, improved in cool_results.items() if improved)

either_improvers   = hot_improvers.union(cool_improvers)
both_improvers     = hot_improvers.intersection(cool_improvers)
hot_only_improvers = hot_improvers.difference(cool_improvers)
cool_only_improvers = cool_improvers.difference(hot_improvers)

print(f"Users who improved in hot weather:      {len(hot_improvers)}")
print(f"Users who improved in cool weather:     {len(cool_improvers)}")
print(f"Users who improved in both:             {len(both_improvers)}")
print(f"Users who improved only in hot weather: {len(hot_only_improvers)}")
print(f"Users who improved only in cool weather:{len(cool_only_improvers)}")
print(f"Users who improved in either:           {len(either_improvers)}")

total_hot_users  = df[df['weather_label'] == 'hot']['user_id'].nunique()
total_cool_users = df[df['weather_label'] == 'cool']['user_id'].nunique()
print(f"\nProportion of hot weather trainers who improved:
{len(hot_improvers)/total_hot_users:.1%}")
print(f"Proportion of cool weather trainers who improved:
{len(cool_improvers)/total_cool_users:.1%}")
```

Users who improved in hot weather: 408

Users who improved in cool weather: 329

Users who improved in both: 314

Users who improved only in hot weather: 94

Users who improved only in cool weather: 15

Users who improved in either: 423

Proportion of hot weather trainers who improved: 95.8%

Proportion of cool weather trainers who improved: 77.8%